ALLIANCE UNIVERSITY | NAAC GRADE A+ ACCREDITED UNIVERSITY

Alliance School of Advanced Computing

Capstone Project – I [7CS1999] [Batch : 2023 – 2027]

Project Title:

MedFedAI: Federated Intelligence for Privacy-Preserving Medical Image Classification

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I. Project Overview

- **Problem Context:** Problems associated with medical image classification include limited training data, class imbalance, resemblance of the diseases, and variance in the data sets. Consequently, CNNs, Transformers, transfer learning, and hybrid methods have been proposed as potential solutions.

- **Motivation:** Manual analysis of medical images is time-consuming and requires expert knowledge. The research aims to develop accurate and robust AI models that can learn meaningful features from medical images.

- **Significance:** Automated medical image analysis can help reduce diagnostic workload and support faster decision-making. It can also improve the consistency and accuracy of disease classification.

- **Application Domain:** The research focuses on AI-based medical image classification and disease detection. It can be applied to X-ray, CT, ultrasound, retinal, skin, and microscopy images.

- **Target Users:** This system is designed mainly for radiologists, medical practitioners, and medical researchers. The system can also serve AI researchers and healthcare organizations that require medical image analysis.

II. Literature Review

S.NO	Author Name Publication Year	Research / Methodology	Performance Metrics	Advantages	Limitations
1	Pan et al., 2026	**MedCTM: A CNN-Transformer-Mamba Hybrid Network for Medical Image Classification**. Combines CNN, Transformer and Mamba branches for local features, long-range dependencies and feature interaction.	Evaluated on **8 medical image datasets** covering 7 modalities and 100,000+ samples. Achieved **1.42–2.63 GFLOPs** and improved average F1 by **0.5–1.7%** over MedViTv2 while using **12.9–24.4% fewer parameters**.	Strong combination of local and global feature learning; better accuracy–efficiency trade-off; relatively lightweight.	More complex architecture because it combines three modelling paradigms; extensive multi-dataset validation is still needed for clinical deployment.
2	Si, Zhang & Zhao, 2026	**A Medical Image Classification Algorithm Based on a Hierarchical and Complementary Attention-Enhanced Swin Transformer Model**. Uses multi-scale, global-context and local-detail enhancement with Swin Transformer.	Tested on **8 MedMNIST v2 subsets**; significant improvements reported on **BreastMNIST, OCTMNIST, OrganSMNIST and PneumoniaMNIST**, with strong robustness and generalization across modalities.	Captures multi-scale lesions and subtle anatomical features; strong cross-modality evaluation.	Still based on Transformer architecture, so computational requirements can remain high; results are mainly benchmark-based rather than clinical validation.
3	Ren et al., 2025	**Conv-SdMLPMixer: A Hybrid Medical Image Classification Network Based on Multi-Branch CNN and Multi-Scale Multi-Dimensional MLP**. Combines multi-branch CNN with a multi-scale MLP feature-fusion module	On **BUSI**, achieved **93.33% accuracy, 93.65% F1, 94.78% precision and 92.66% recall**. Also evaluated on COVID19-CT, Small-ISIC2018 and Chest-Xray.	Captures local lesion details and broader spatial/channel information; performs well on small-lesion and noisy datasets.	Hybrid design increases architectural complexity; performance may depend on dataset characteristics and requires broader external validation.
4	MedSetFeat++ authors, 2026	**MedSetFeat++: An Attention-Enriched Set Feature Framework for Few-Shot Medical Image Classification**. Uses multi-head/multi-scale attention, positional embeddings, adaptive head gating and CBAM.	Under **2-way 10-shot 15-query**, achieved **92.17% accuracy on HAM10000, 70.89% on BreakHis and 73.46% on Kvasir**.	Useful for **limited labelled data**; strong few-shot performance; combines spatial, semantic and contextual features.	Few-shot results are task/setup dependent; evaluation is limited to selected datasets and episode settings.
5	Asimeng et al., 2026	**Transformer-Masked Autoencoder (MAE) for Robust Medical Image Classification: A Comprehensive Survey**. Reviews MAE-based approaches for X-ray, MRI, CT, ultrasound and histopathology.	Survey of **110 studies**; reports that MAE initialization improves label efficiency, calibration and cross-scanner transfer, especially with parameter-efficient fine-tuning.	Very useful for identifying current trends in **self-supervised learning and label-efficient classification**.	It is a **survey**, not a new classification model; reported studies use different datasets and evaluation protocols, making direct comparison difficult.

II. Literature Review

S.NO	Author Name Publication Year	Research / Methodology	Performance Metrics	Advantages	Limitations
6	Zheng et al., 2024	Proposed **MedPrompt**, a weakly supervised prompt-learning approach for low-resource medical image classification.	MedPrompt-ViT achieved **AUC 0.658 (CheXpert), 0.616 (MIMIC-CXR), 0.9553 (COVID), 0.8304 (RSNA)** in full-shot evaluation.	Reduces dependence on manually designed expert prompts; supports zero-shot and few-shot classification.	Prompt wording can strongly affect performance; domain-expert dependence is reduced but not completely eliminated.
7	Liu et al., 2024	Proposed **Eff-CTNet**, an efficient alternating CNN–Transformer architecture with Group Cascade Attention.	On COVID19-CT: **92.57% accuracy, 93.17% F1, 91.46% precision, 94.94% recall, AUC 0.9240**.	Efficiently combines local and global features while reducing attention redundancy and computational cost.	Slight overfitting caused by noise and class imbalance was observed on Chaoyang.
8	Wu et al., 2023	Proposed **CTransCNN**, combining CNN and Transformer branches for multilabel medical image classification.	Average AUC **83.37%** on ChestX-ray11 and **78.47%** on NIH ChestX-ray14; AUC **84.56%** and mAP **67.67%** on TCMTD.	Models local/global features, label correlations and class imbalance jointly.	Severe label imbalance; some datasets have limited samples and the hybrid architecture increases computational complexity.
9	Manzari et al., 2023	Proposed **MedViT**, a hybrid CNN–Vision Transformer for generalized medical image classification.	Average AUC: **93.6%, 94.2%, 93.5%** and average accuracy: **84.0%, 85.1%, 84.2%** for MedViT-T, S and L.	Combines CNN local features with Transformer global features; good accuracy–complexity trade-off.	Transformers require substantial training data and computation; medical images have large texture, shape and size variations.
10	Wang et al. / MedFMC, 2023	Introduced **MedFMC**, a benchmark for foundation-model adaptation and few-shot medical image classification across multiple modalities.	Fine-tuned DenseNet-121, EfficientNet-B4 and Swin-Base; performance evaluated using **Accuracy, AUC and mAP** across five datasets.	Supports few-shot and transfer-learning evaluation across diverse medical tasks and modalities.	Performance drops with very few patient samples; head–tail class imbalance remains a challenge.

II. Literature Review

S.NO	Author Name Publication Year	Research / Methodology	Performance Metrics	Advantages	Limitations
11	Kim et al., 2022	Conducted a literature review of **transfer learning with pretrained CNNs** for medical image classification.	Reviewed **425 articles**, with **121 studies** meeting the inclusion criteria; Inception was the most frequently used deep model.	Shows transfer learning is effective with limited medical data; provides recommendations for selecting CNN backbones and TL strategies.	Focused mainly on pretrained CNNs; Vision Transformers and source–target domain differences were not evaluated
12	Huynh, Nibali & He, 2021	Proposed **Adaptive Blended Consistency Loss (ABCL)** for semi-supervised medical image classification with class imbalance.	On HAM10000, UDA-WeightedCE-ABCL achieved **UAR 0.68, G-mean 0.66, average AUC 0.96**.	Handles both limited labels and class imbalance; improves minority-class performance.	Severe class imbalance remains challenging; strong augmentation can reduce the effectiveness of ABCL.
13	Azizi et al., 2021	Studied **self-supervised pretraining** with SimCLR and proposed **MICLe** for medical image classification.	**+6.7% top-1 accuracy** in dermatology and **+1.1% mean AUC** in chest X-ray classification.	Reduces dependence on labelled data; improves robustness to distribution shift and label efficiency.	Requires suitable large-scale unlabeled data and substantial computational resources; domain differences between natural and medical images remain important.
14	Yadav & Jadhav, 2019	Compared **ORB+SVM, VGG16 transfer learning, InceptionV3 transfer learning and Capsule Network** for pneumonia classification from chest X-rays.	Transfer learning was the best of the three main approaches; VGG16 with one retrained convolution layer gave the best result.	Effective for small datasets; data augmentation improves performance.	Fine-tuning can overfit; model complexity must match dataset size; visualization and stronger CNNs were suggested as future work.
15	Smitha Rajagopal et al., 2025	FL + Deep CapsNet for brain-tumor MRI classification with U-Net feature extraction.	96.79% Accuracy, 96.84% Recall, 96.85% F1, 0.97 AUC	Privacy-preserving, decentralized training, good classification performance.	High computational cost, limited scalability, Non-IID handling needs improvement.

III. Research Gap

1. Identified Research Gap:

Existing research has produced accurate results for medical image classification tasks; however, issues such as insufficient labeled training data, imbalanced classes, domain shift, and high costs associated with computations still exist. It is therefore important to have a solution that is not only robust but also computationally efficient and generalizable across medical data.

2. Technical Challenges:

- Non-Independent and Identically distributed (Non-IID) data across healthcare clients can reduce global-model performance.
- Diverse patient populations, modalities and acquisition conditions create distribution shift.
- Severe class imbalance can bias the classifier toward majority classes.
- Repeated model communication introduces communication and computational overhead.
- Privacy mechanisms may create accuracy and computation trade-offs.
- Broader multi-institutional validation is still a major concern.

3. Need for Proposed Work:

MedFedAI aims to provide a robust framework to test whether Federated Learning achieves accurate medical image classification while preserving patient privacy, especially when data across hospitals is different (Non-IID).

IV. Requirements Analysis

Software Requirements

- Python → project implementation
- TensorFlow / PyTorch → deep-learning model development
- NumPy & Pandas → data processing
- Scikit-learn → preprocessing and evaluation
- Flower / TensorFlow Federated → Federated Learning simulation
- Matplotlib → result visualization
- Google Colab / Jupyter → experiments
- Git / GitHub → version control

Dataset / Resource Requirements

- Medical image datasets from Multiple clients/hospitals.
- Both IID and Non-IID data distributions should be considered for evaluation.
- Sufficient labelled and unlabelled images for training and testing. Computing resources (GPU/CPU, storage)
- Multiple client environments to simulate different healthcare institutions.
- **Example Datasets**: Brain tumour, Skin lesions, ultrasound, chest X-ray, endoscopy, pathology/retinal/etc.

IV. Requirements Analysis

Constraints:

- Medical data is sensitive and privacy-protected, so raw data cannot be directly shared between institutions.
- Non-IID data distribution can affect the performance of the federated model.
- Limited computational resources may restrict the use of complex models.
- Communication between participating clients can increase training time and overhead.

Proposed Differentiation:

The proposed MedFedAI framework differs by combining privacy-preserving Federated Learning with medical image classification, while evaluating the performance using heterogeneous (IID and realistic Non-IID) data settings. It aims to provide robust and reliable classification without requiring sensitive medical data to be shared between the institutions.

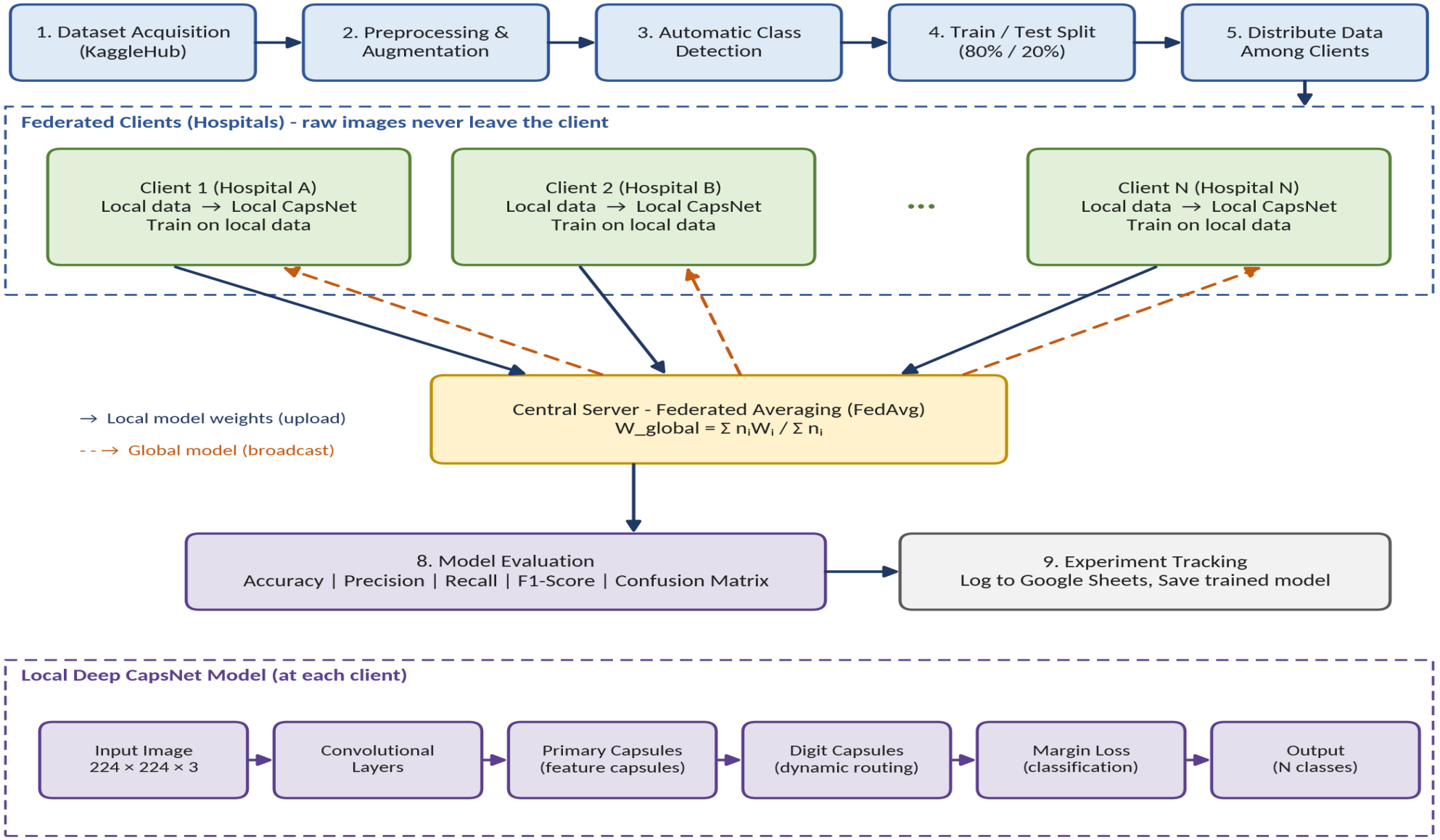
V. Problem Statement

Medical image classification is challenged by limited labelled data, class imbalance, variations across healthcare institutions, and patient privacy concerns, **necessitating a federated learning framework using Deep Capsule Networks for accurate and privacy-preserving collaborative classification without sharing raw medical images.**

VI. Objectives

1. Develop a privacy-preserving federated learning framework using Deep Capsule Network (Deep CapsNet) for medical image classification, enabling multiple clients to collaboratively train a model without sharing raw medical images.
2. Test classification accuracy and generalization by addressing limited labeled data, class imbalance, and variations in medical images across different clients through appropriate preprocessing, data augmentation, and balanced training strategies.
3. Evaluate the performance and reliability of the proposed framework using accuracy, precision, recall, F1-score, and confusion matrix, while maintaining data privacy during federated training.

VII. Architecture



VIII. Proposed Methodology stepwise

1. **Dataset Acquisition:** Collect labelled medical images and organize them into disease classes.
2. **Preprocessing:** Resize images to 224 × 224, normalize pixel values, apply augmentation, and handle class imbalance.
3. **Client Distribution:** Split the dataset into training and testing sets and distribute training data across simulated hospital clients under IID and Non-IID settings.
4. **Local Model Training:** Train a Deep Capsule Network independently at each client using convolutional layers, primary capsules, dynamic routing, and margin loss.
5. **Federated Aggregation:** Use Federated Averaging **(FedAvg)** to combine local model updates at a central server without transferring raw medical images.
6. **Global Model Update:** Broadcast the aggregated model to clients and repeat training across multiple communication rounds.
7. **Performance Evaluation:** Measure **accuracy, precision, recall, F1-score, confusion matrix, convergence, and communication overhead.**
8. **Privacy Analysis:** Assess the privacy benefits and limitations of decentralized training and investigate additional privacy mechanisms where necessary.

IX. Implementation Results

1. Implementation :

https://colab.research.google.com/drive/1Ei5QddO2s0S1gP702YYozlw3__ZABXhU

2. Results:

Parameter	Chest X-ray Pneumonia	Brain Tumor MRI	Digital Knee X-ray
Model	Federated Deep CapsNet	Federated Deep CapsNet	Federated Deep CapsNet
Image Size	128 × 128	128 × 128	128 × 128
Classes	2	4	2
Clients	4	4	4
Fed. Rounds	3	3	3
Local Epochs	1	1	1
Batch Size	4	4	4
Learning Rate	0.0005	0.0005	0.0005
Training Images	1,000	2,000	800
Testing Images	624	1,600	200
Accuracy (%)	67.95	56.06	46.50
Precision (%)	74.13	60.76	44.63
Recall (%)	67.95	56.06	46.50
F1 Score (%)	60.30	51.43	41.40
Test Loss	0.2081	0.2645	0.2773

X. References

[1]Pan, W., Kang, J., Wang, X., Lv, C., Zhang, X., & Liu, X. (2026). MedCTM: a CNN-transformer-mamba hybrid network for medical image classification. *Information Processing & Management*, *63*(7), 104867.

[2]Si, Y., Zhang, Y., & Zhao, M. (2026). A medical image classification algorithm based on a hierarchical and complementary attention-enhanced Swin Transformer model. *Scientific Reports*.

[3]Ren, Z., Liu, S., Wang, L., & Guo, Z. (2025). Conv-SdMLPMixer: A hybrid medical image classification network based on multi-branch CNN and multi-scale multi-dimensional MLP. *Information Fusion*, *118*, 102937.

[4]Titoriya, A. K., Singh, M. P., & Singh, A. K. (2025). MedSetFeat++: An attention-enriched set feature framework for few-shot medical image classification. *Image and Vision Computing*, 105825.

[5]Asimeng, E., Chen, J., Han, K., Lyu, C., & Liu, Z. (2026). Transformer-masked autoencoder (MAE) for robust medical image classification: A comprehensive survey. *Expert Systems with Applications*, 131462.

[6]Zheng, F., Cao, J., Yu, W., Chen, Z., Xiao, N., & Lu, Y. (2024). Exploring low-resource medical image classification with weakly supervised prompt learning. *Pattern Recognition*, *149*, 110250.

[7]Liu, S., Yue, W., Guo, Z., & Wang, L. (2024). Multi-branch CNN and grouping cascade attention for medical image classification. *Scientific Reports*, *14*(1), 15013.

[8]Wu, X., Feng, Y., Xu, H., Lin, Z., Chen, T., Li, S., ... & Zhang, S. (2023). CTransCNN: Combining transformer and CNN in multilabel medical image classification. *Knowledge-Based Systems*, *281*, 111030.

X. References

[9]Manzari, O. N., Ahmadabadi, H., Kashiani, H., Shokouhi, S. B., & Ayatollahi, A. (2023). MedViT: a robust vision transformer for generalized medical image classification. *Computers in biology and medicine*, *157*, 106791.

[10]Wang, D., Wang, X., Wang, L., Li, M., Da, Q., Liu, X., ... & Zhang, S. (2023). A real-world dataset and benchmark for foundation model adaptation in medical image classification. *Scientific Data*, *10*(1), 574.

[11]Kim, H. E., Cosa-Linan, A., Santhanam, N., Jannesari, M., Maros, M. E., & Ganslandt, T. (2022). Transfer learning for medical image classification: a literature review. *BMC medical imaging*, *22*(1), 69.

[12]Huynh, T., Nibali, A., & He, Z. (2022). Semi-supervised learning for medical image classification using imbalanced training data. *Computer methods and programs in biomedicine*, *216*, 106628.

[13]Azizi, S., Mustafa, B., Ryan, F., Beaver, Z., Freyberg, J., Deaton, J., ... & Norouzi, M. (2021, October). Big self-supervised models advance medical image classification. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)* (pp. 3458-3468). IEEE.

[14]Yadav, S. S., & Jadhav, S. M. (2019). Deep convolutional neural network based medical image classification for disease diagnosis. *Journal of Big data*, *6*(1), 1-18.

[15] Rajagopal, S., Shieh, C. S., Shankar, S. S., Maithili, K., & Chakrabarti, P. (2026). Privacy-preserving multi-class brain tumor classification using federated learning and deep capsule networks. *Soft Computing*, *30*(4), 2769-2787.

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Thank you

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